



Technical Service Bulletin

Technical Service Bulletin: TSB120214

Released Date: 06-Jul-2012

Turbocharger Impeller Supersession Impacts HX82, HE800, and HX80 Turbocharger Repairs

Turbocharger Impeller Supersession Impacts HX82, HE800, and HX80 Turbocharger Repairs

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

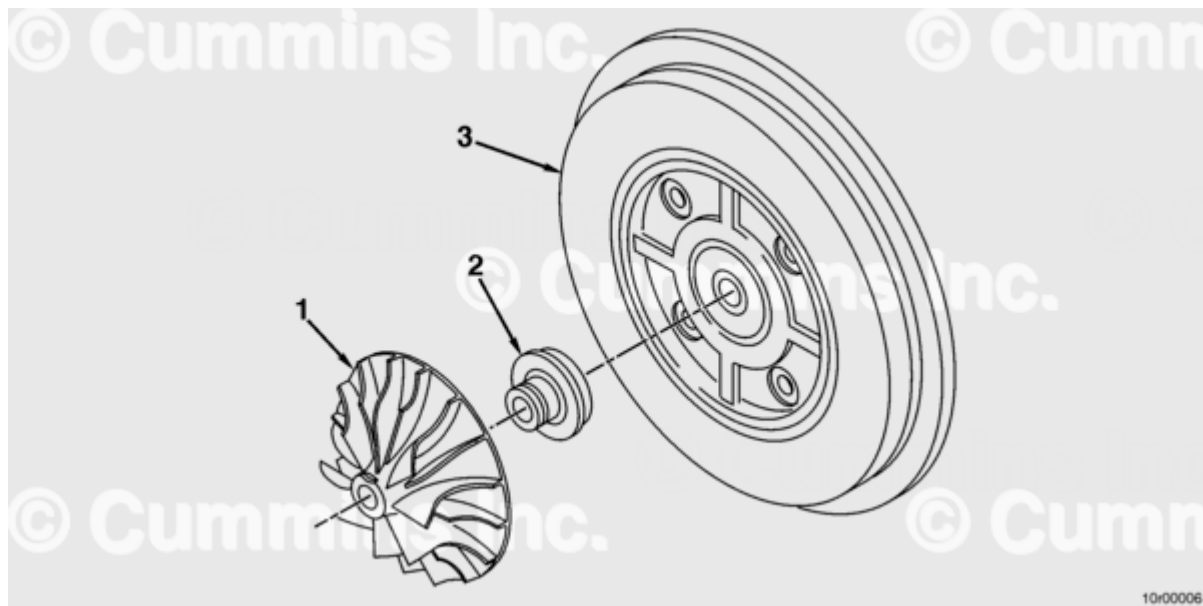


Figure 1; Parts affected in the impeller supersession: Impeller (1), Oil Slinger (2), and Diffuser (3).

A change was made to the impeller wheel used on HX82, HE800, and HX80 turbocharger models. This change affects the parts needed for turbocharger repairs.

When replacing a turbocharger rotor assembly or impeller wheel (1) on HX82, HE800FG, and HX80 turbochargers, the impeller part number **must** be checked. The affected impeller part numbers **must** use a corresponding new oil slinger (2) and new diffuser plate (3) for proper assembly.

The affected impeller part numbers are:

- 4048018

- 4048019
- 4048020
- 4048021
- 4048022
- 2839630
- 2839631
- 2839632

If a turbocharger is being repaired with any of the impellers listed above, the diffuser plate part number **must** be checked. Reference the newest turbocharger part number supersession for the correct diffuser part number. If the diffuser part number in the most recent supersession does **not** match the diffuser part number of the turbocharger being repaired, the newest diffuser plate and the oil slinger, Part Number 4036307, **must** be ordered. The overhaul kit (Bearing Repair Kit) will contain new capscrews and washers to mount the replacement diffuser plate. The overhaul kit will also contain a new seal ring for the oil slinger.

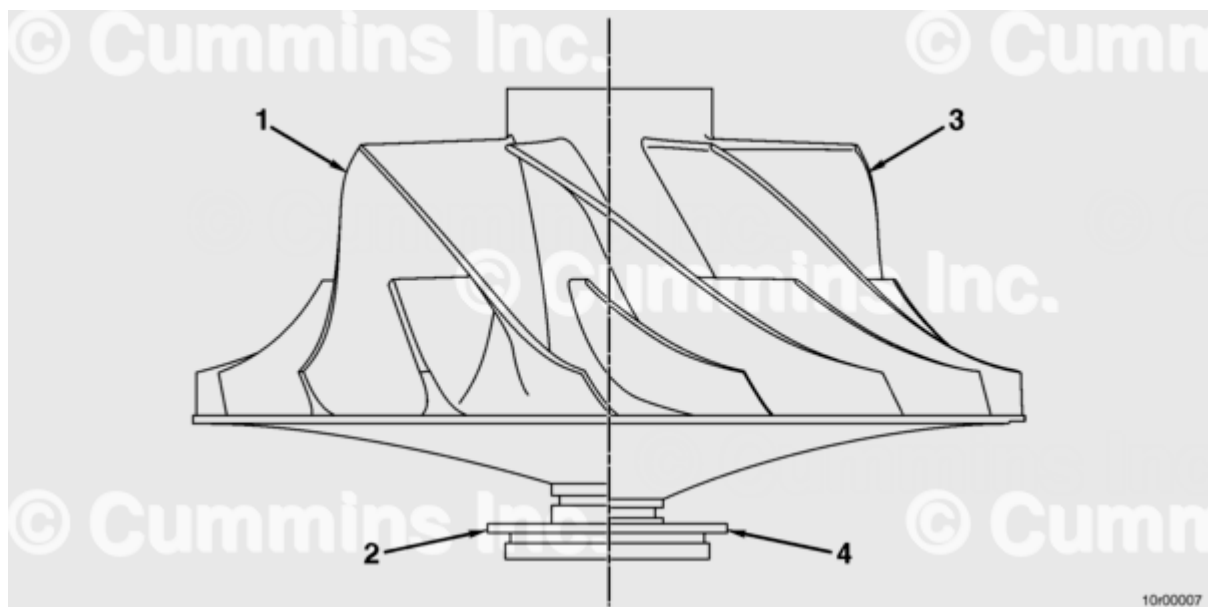


Figure 2; Comparison of Old Impeller Wheel (1) / Oil Slinger (2) and New Impeller Wheel (3) / Oil Slinger (4).

The new impeller (3) in Figure 2 has a longer back face, which requires a different diffuser plate and oil slinger (4).

Note : If a turbine wheel and/or impeller are bought as a piece part for a turbocharger rebuild, the rotor assembly (turbine shaft/wheel and impeller) **must** be balanced. Core assemblies and rotor assemblies are balanced from the manufacturer. For rotor assemblies, make sure the impeller and shaft are aligned as received.

Note : The diffuser capscrews are patch treated and new capscrews **must** be used when installing the diffuser plate.

Use Table 1 to look up part number supersessions.

Table 1; Part Number Look Up Table for Supersessions						
ID17 (Turbocharger Part Number)	Old Impeller	Old Diffuser	Old Slinger	New Impeller	New Diffuser	New Slinger
3593549	3593923	3534703	3523980	4048018	4048023	4036307
3593986	3593923	3534703	3523980	4048018	4048023	4036307
3591398	3593924	3534703	3523980	4048019	4048023	4036307
3592855	3593924	3534703	3523980	4048019	4048023	4036307
3593552	3593924	3534703	3523980	4048019	4048023	4036307
3596024	3593924	3534703	3523980	4048019	4048023	4036307
3593232	3593925	3534703	3523980	4048020	4048023	4036307
3595847	3593925	3534703	3523980	4048020	4048023	4036307
4041702	3593925	4039429	3523980	4048020	2838966	4036307
4047475	3593925	4039429	3523980	4048020	2838966	4036307
4048353	3593925	4039429	3523980	4048020	2838966	4036307
2834699	3593926	4039429	3523980	4048021	2838966	4036307
2835774	3593926	3534703	3523980	4048021	4048023	4036307
3591403	3593926	3534703	3523980	4048021	4048023	4036307
3594181	3593926	3534703	3523980	4048021	4048023	4036307
3594587	3593926	3534703	3523980	4048021	4048023	4036307
3595850	3593926	3534703	3523980	4048021	4048023	4036307
3599340	3593926	3599931	3523980	4048021	3767445	4036307
3599346	3593926	3599931	3523980	4048021	3767445	4036307
4035682	3593926	3534703	3523980	4048021	4048023	4036307
4036075	3593926	3534703	3523980	4048021	4048023	4036307
4040703	3593926	3534703	3523980	4048021	4048023	4036307
4040749	3593926	4039429	3523980	4048021	2838966	4036307
4042384	3593926	3534703	3523980	4048021	4048023	4036307
4042798	3593926	4039429	3523980	4048021	2838966	4036307
4043346	3593926	4039429	3523980	4048021	2838966	4036307
4043668	3593926	3534703	3523980	4048021	4048023	4036307
4044033	3593926	4039429	3523980	4048021	2838966	4036307
4044415	3593926	3534703	3523980	4048021	4048023	4036307
4044551	3593926	4039429	3523980	4048021	2838966	4036307
4047173	3593926	3534703	3523980	4048021	4048023	4036307
4049905	3593926	4039429	3523980	4048021	2838966	4036307
3594184	3593927	3534703	3523980	4048022	4048023	4036307

Table 1; Part Number Look Up Table for Supersessions						
ID17 (Turbocharger Part Number)	Old Impeller	Old Diffuser	Old Slinger	New Impeller	New Diffuser	New Slinger
3594441	3593927	3534703	3523980	4048022	4048023	4036307
3597614	3593927	3534703	3523980	4048022	4048023	4036307
2840120	3533704	3527104	3523980	2839630	2841902	4036307
2840584	3533705	3592131	3523980	2839632	3767933	4036307
2840585	3533705	3592131	3523980	2839632	3767933	4036307
3527372	3533703	3527348	3523980	2839631	3767932	4036307
3527709	3533705	3527348	3523980	2839632	3767932	4036307
3528460	3533704	3527104	3523980	2839630	2841902	4036307
3529870	3533703	3527348	3523980	2839631	3767932	4036307
3530185	3533703	3527348	3523980	2839631	3767932	4036307
3531530	3533703	3527348	3523980	2839631	3767932	4036307
3531983	3533703	3527348	3523980	2839631	3767932	4036307
3532932	3533703	3527348	3523980	2839631	3767932	4036307
3532946	3533703	3527348	3523980	2839631	3767932	4036307
3533427	3533703	3527348	3523980	2839631	3767932	4036307
3533723	3533703	3527348	3523980	2839631	3767932	4036307
3534120	3533703	3527348	3523980	2839631	3767932	4036307
3534243	3533703	3527348	3523980	2839631	3767932	4036307
3534562	3533703	3527348	3523980	2839631	3767932	4036307
3534625	3533703	3527348	3523980	2839631	3767932	4036307
3534645	3533703	3527348	3523980	2839631	3767932	4036307
3536135	3533703	3527348	3523980	2839631	3767932	4036307
3536363	3533703	3527348	3523980	2839631	3767932	4036307
3536366	3533703	3527348	3523980	2839631	3767932	4036307
3537102	3533704	3527104	3523980	2839630	2841902	4036307
3537683	3533703	3538455	3523980	2839631	3767931	4036307
3537685	3533703	3538455	3523980	2839631	3767931	4036307
3537921	3533704	3527104	3523980	2839630	2841902	4036307
3538266	3533705	3527348	3523980	2839632	3767932	4036307
3538512	3533703	3538455	3523980	2839631	3767931	4036307
3538513	3533703	3538455	3523980	2839631	3767931	4036307
3538610	3533703	3527348	3523980	2839631	3767932	4036307
3539278	3533704	3527104	3523980	2839630	2841902	4036307

Table 1; Part Number Look Up Table for Supersessions						
ID17 (Turbocharger Part Number)	Old Impeller	Old Diffuser	Old Slinger	New Impeller	New Diffuser	New Slinger
3539284	3533704	3527104	3523980	2839630	2841902	4036307
3539287	3533704	3527104	3523980	2839630	2841902	4036307
3590178	3533703	3538455	3523980	2839631	3767931	4036307
3590196	3533703	3538455	3523980	2839631	3767931	4036307
3591748	3533705	3592131	3523980	2839632	3767933	4036307
3594096	3533703	3527348	3523980	2839631	3767932	4036307
3594117	3533703	3527348	3523980	2839631	3767932	4036307
3594126	3533703	3527348	3523980	2839631	3767932	4036307
3594128	3533703	3527348	3523980	2839631	3767932	4036307
3594131	3533703	3527348	3523980	2839631	3767932	4036307
3594137	3533703	3527348	3523980	2839631	3767932	4036307
3594139	3533703	3538455	3523980	2839631	3767931	4036307
3594141	3533703	3538455	3523980	2839631	3767931	4036307
3594147	3533705	3527348	3523980	2839632	3767932	4036307
3594149	3533703	3538455	3523980	2839631	3767931	4036307
3594151	3533703	3538455	3523980	2839631	3767931	4036307
3594153	3533703	3527348	3523980	2839631	3767932	4036307
3594157	3533704	3527104	3523980	2839630	2841902	4036307
3594163	3533704	3527104	3523980	2839630	2841902	4036307
3594166	3533704	3527104	3523980	2839630	2841902	4036307
3594172	3533703	3538455	3523980	2839631	3767931	4036307
3594174	3533703	3538455	3523980	2839631	3767931	4036307
3596154	3533704	3527104	3523980	2839630	2841902	4036307
3596959	3533703	3538455	3523980	2839631	3767931	4036307
3596961	3533703	3538455	3523980	2839631	3767931	4036307
3596963	3533703	3538455	3523980	2839631	3767931	4036307
3597927	3533705	3592131	3523980	2839632	3767933	4036307
3599005	3533703	3527348	3523980	2839631	3767932	4036307
4037724	3533703	3527348	3523980	2839631	3767932	4036307
4037934	3533705	3527348	3523980	2839632	3767932	4036307
4038942	3533703	3527348	3523980	2839631	3767932	4036307
4040699	3533705	3592131	3523980	2839632	3767933	4036307
4041143	3533704	3527104	3523980	2839630	2841902	4036307

Table 1; Part Number Look Up Table for Supersessions						
ID17 (Turbocharger Part Number)	Old Impeller	Old Diffuser	Old Slinger	New Impeller	New Diffuser	New Slinger
4042381	3533705	3592131	3523980	2839632	3767933	4036307
4044402	3533704	3527104	3523980	2839630	2841902	4036307
4044421	3533703	3527348	3523980	2839631	3767932	4036307
4044427	3533704	3527104	3523980	2839630	2841902	4036307
4044770	3533705	3592131	3523980	2839632	3767933	4036307
4046454	3533703	3527348	3523980	2839631	3767932	4036307

Document History

Date	Details
2012-6-27	Module Created